INCH-POUND

MIL-PRF-1/812D
17 September 1999
SUPERSEDING
MIL-E-1/812C
22 May 1975

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 5D22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Tetrode.

See figure 1.

Mounting position: Vertical, base up or down.

Weight: 8 ounces (226.8 grams) nominal.

ABSOLUTE RATINGS: C Telegraphy

Parameter:	F1	Ef	Eb	Ec1	Ec2	Ib	Pp	Pg1	Pg2	Cooling
Unit:	MHz	V ac	V dc	V dc	V dc	mA dc	W	W	W	1/
Maximum:	75	5.25	4,000	-500	600	350	250	10	35	---
Minimum:	---	4.75	---	---	---	---	---	---	---	---
Test conditions:	---	5.0	2,500	Adjust	500	100	---	---	---	2/

See footnotes at end of table I.

GENERAL:

Qualification: Required.

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Acceptance Level 5/	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 1								
Filament current	1301	---		0.65	If	13.5	14.7	A ac
Total grid current	1266	3/		0.65	Ic1	---	-10	µA dc
Primary grid emission (control)	1266	---	Ef = 5.5 V ac; Ic1 = 165 mA dc; t = 15 sec; anode and screen grid floating	0.65	Isg1	---	-100	µA dc
Primary grid emission (screen)	1266	---	Ef = 5.5 V ac; Ec1 = 0; Ic2 = 135 mA dc; t = 15 sec; anode floating	0.65	Isg2	---	-200	µA dc
Electrode voltage (grid)	1261	---		0.65	Ec1	-65	-95	V dc
Peak emission	1231	---	eb = ec1 = ec2 = 2,500 v	0.65	is	5.0	---	a
Conformance inspection, part 2								
Amplification factor	1316	---	g1 to g2; Ec2 = 500 V dc; Ic2 = 70 mA dc; Eb = 0 (anode grounded)	---	Mu	4.5	6.0	---
Power oscillation (1)	1236	---	Ebb = 3,000 V dc; Ib = 200 mA dc; Ec2 = 350 V dc; Rg1 = 12,000 ohms; F = 15 MHz	--- ---	Po Ic2	350 50	--- 100	W (useful) mA dc
Direct-interelectrode capacitance	1331	---		--- --- ---	Cgp Cin Cout	--- 10.7 3.7	0.14 14.5 5.1	pF pF pF

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Notes	Conditions	Acceptance Level 5/	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 3								
Low-frequency vibration	1031	4/	No voltages applied	---	---	---	---	---
Bump	1036	4/	Angle = 20°	---	---	---	---	---
Low-frequency vibration and shock end points:	---							
Electrode voltage (grid)	1261	---		---	Ec1	-65	-95	V dc
Total grid current	1266	---		---	Ic1	---	-10	µA dc
Life test	---	---	Group C; power oscillation (1); t = 500 hours	---	---	---	---	---
Life-test end points:	---							
Peak emission	1231	---		---	is	4.0	---	a
Primary grid emission (control)	1266	---		---	Isg1	---	-100	µA dc
Primary grid emission (screen)	1266	---		---	Isg2	---	-200	µA dc
Total grid current	1266	---		---	Ic1	---	-10	µA dc
Power oscillation (2)	1236	4/	Ebb = 3,000 V dc; Ib = 200 mA dc; Ec2 = 350 V dc; Rg1 = 12,000 ohms; F = 75 MHz	---	Po	350	---	W (useful)

1/ Adequate cooling shall be provided for the seals and envelope of the tube. An air-system socket and air director (chminey) is desirable, such as Eimac SK-400/SK-406 or SK-410/SK-406, or equal. At frequencies below 30 MHz, cooling air in the amount of 5 cfm around the base of the tube past the envelope is required. At frequencies above 30 MHz, losses in the leads and envelope contribute to heating, and additional cooling may be required. Cooling should be applied simultaneously with filament power, and in all cases there should be some movement of air past the base and envelope.

2/ Cooling of the tube is permitted to an extent not to exceed that specified in 1/.

3/ This test shall be the first test performed at the conclusion of the holding period.

4/ This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of zero. In the event of failure, the test will be made as a part of conformance inspection, part 2, code level D, with an acceptance level of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

5/ This specification uses accept on zero (c = 0) sampling plan, in accordance with MIL-PRF-1, Table III.

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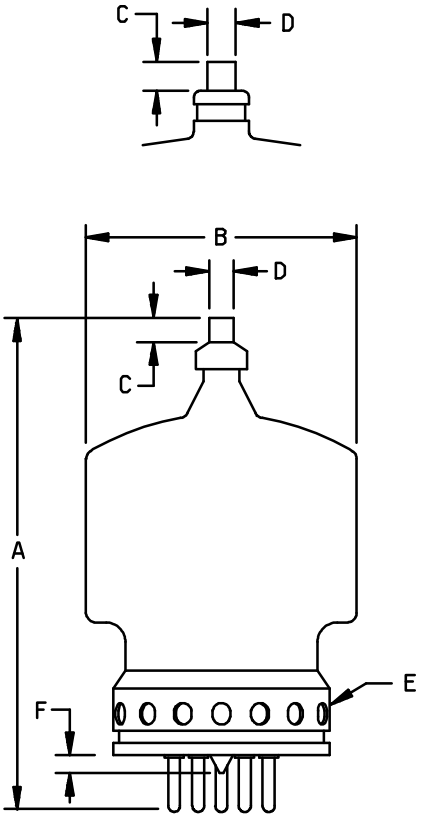

Pin connections	
Pin no.	Element
1	f
2	g2
3	g1
4	g2
5	f
cap	a

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	5.875	6.375	149.23	161.93
B	---	3.563	---	90.50
F	---	.250	---	6.35
Conformance inspection, part 3 (see note)				
C	.328	---	8.33	---
D	.350	.365	8.89	9.27
E	A5-97 (EIA)			

NOTE:
These dimensions shall be checked annually with the following sampling plan:

n1 = 4 c1 = 0 where c2 represents the total failures for the
n2 = 4 c2 = 0; first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become conformance inspection, part 2, for three successful consecutive submissions, at which time the test may revert to the conformance inspection, part 3, basis.

FIGURE 1. Outline drawing of electron tube type 5D22.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5960-3558-03)

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